

LP5 Practical Viva Preparation Guide

PART A: HIGH PERFORMANCE COMPUTING (HPC)

1. What is High Performance Computing (HPC)?

Detailed Explanation

High Performance Computing means using multiple processors, multiple CPU cores, or GPUs to solve large and complex problems faster.

Normally, a single CPU performs one task at a time. In HPC, the problem is divided into smaller parts, and many processors work on these parts simultaneously.

HPC is used in:

- Weather forecasting
- Scientific simulations
- Artificial Intelligence
- Medical research
- Space exploration

Real-Life Example

If one student solves 100 math problems alone, it takes a long time. If 100 students solve one problem each, the work finishes much faster.

Medium Answer

HPC uses multiple processors or GPUs to solve large computational problems quickly by executing many tasks in parallel.

One-Line Answer

HPC is the use of parallel processing to solve complex problems faster.

2. Sequential vs Parallel Computing

Detailed Explanation

Sequential Computing

One instruction executes after another.

Parallel Computing

Multiple instructions execute simultaneously.

Real-Life Example

Washing 100 plates:

- Sequential: one person washes all plates.
- Parallel: 10 people wash plates together.

Medium Answer

Sequential computing executes tasks one by one, while parallel computing executes multiple tasks at the same time.

One-Line Answer

Parallel computing performs many tasks simultaneously.

3. What is a Thread?

Detailed Explanation

A thread is the smallest unit of execution inside a program.

One program can create multiple threads. Each thread performs part of the work.

Real-Life Example

A restaurant has many workers. Each worker handles one task. Each worker is like a thread.

One-Line Answer

A thread is a lightweight execution unit.

4. What is OpenMP?

Detailed Explanation

OpenMP is a library used in C and C++ to write parallel programs for multicore CPUs.

We use compiler directives like:

```
#pragma omp parallel for
```

This tells the compiler to distribute loop iterations among multiple threads.

Real-Life Example

A teacher divides homework among many students.

Medium Answer

OpenMP simplifies parallel programming by automatically creating threads and distributing work.

One-Line Answer

OpenMP is an API for parallel programming on CPU.

5. What is CUDA?

Detailed Explanation

CUDA (Compute Unified Device Architecture) is a technology developed by NVIDIA that allows C/C++ code to run on the GPU.

GPUs contain thousands of cores, making them ideal for tasks where the same operation is repeated on many data elements.

Real-Life Example

Instead of one worker doing all additions, thousands of workers each perform one addition.

Medium Answer

CUDA enables programs to use the GPU for massive parallel processing.

One-Line Answer

CUDA is a GPU programming platform developed by NVIDIA.

6. CPU vs GPU

CPU	GPU
Few powerful cores	Thousands of smaller cores
Good for general tasks	Good for highly parallel tasks
Sequential processing	Massive parallel processing

One-Line Answer

CPU has few powerful cores, while GPU has thousands of cores for parallel tasks.

7. What is Speedup?

Detailed Explanation

Speedup tells how much faster the parallel version is compared to the sequential version.

Formula: $\text{Speedup} = \text{Sequential Time} / \text{Parallel Time}$

Example

Sequential = 10 ms Parallel = 2 ms Speedup = 5x

One-Line Answer

Speedup is the ratio of sequential time to parallel time.

8. What is Time Complexity?

Detailed Explanation

Time complexity describes how execution time grows as input size increases.

Examples:

- $O(1) \rightarrow$ constant
- $O(n) \rightarrow$ linear
- $O(n \log n) \rightarrow$ efficient sorting
- $O(n^2) \rightarrow$ slower for large data

One-Line Answer

Time complexity shows how runtime grows with input size.

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Experiment 1: Parallel BFS

What is BFS?

BFS (Breadth First Search) visits nodes level by level.

Data Structure Used

Queue (FIFO)

Real-Life Example

Find all your direct friends first, then friends of friends.

Time Complexity

$O(V + E)$

Space Complexity

$O(V + E)$

Why Use Visited Array?

To prevent revisiting nodes.

Can BFS Find Shortest Path?

Yes, in unweighted graphs.

One-Line Answer

BFS uses a queue to visit nodes level by level.

Experiment 2: Parallel DFS

What is DFS?

DFS (Depth First Search) explores one path completely before backtracking.

Data Structure Used

Stack

Real-Life Example

Explore one branch of a maze until dead end, then return.

Time Complexity

$O(V + E)$

Space Complexity

$O(V + E)$

One-Line Answer

DFS uses a stack to explore nodes deeply before backtracking.

Experiment 3: Parallel Bubble Sort

What is Bubble Sort?

Repeatedly compares adjacent elements and swaps them if they are in the wrong order.

Why Called Bubble Sort?

Largest element moves to the end like a bubble rising.

Parallel Version

Uses Odd-Even Transposition Sort.

Time Complexity

$O(n^2)$

Space Complexity

$O(1)$

Stable?

Yes.

One-Line Answer

Bubble Sort repeatedly swaps adjacent elements until the array is sorted.

Experiment 4: Parallel Merge Sort

What is Merge Sort?

A divide-and-conquer algorithm.

Steps

1. Divide array into two halves.
2. Sort each half.

3. Merge sorted halves.

Time Complexity

$O(n \log n)$

Space Complexity

$O(n)$

Stable?

Yes.

Why Better Than Bubble Sort?

Much faster for large arrays.

One-Line Answer

Merge Sort divides the array, sorts each half, and merges them.

Experiment 5: Parallel Reduction

What is Reduction?

Combining many values into one.

Operations

- Minimum
- Maximum
- Sum
- Average

Time Complexity

$O(n)$

Space Complexity

$O(1)$

One-Line Answer

Reduction combines multiple values into one final result.

Experiment 6: CUDA Vector Addition

What is a Vector?

A one-dimensional array.

What Does Each Thread Do?

Adds one pair of elements.

Time Complexity

$O(n)$

Space Complexity

$O(n)$

One-Line Answer

Each CUDA thread adds one element of the two vectors.

Experiment 7: CUDA Matrix Multiplication

What Does Each Thread Do?

Computes one element of the result matrix.

Formula

$$C[i][j] = \sum A[i][k] \times B[k][j]$$

Time Complexity

$O(n^3)$

Space Complexity

$O(n^2)$

One-Line Answer

Each CUDA thread computes one element of the result matrix.

Important OpenMP Questions

What is `#pragma omp parallel` for?

It distributes loop iterations among multiple threads.

What is `omp_get_wtime()`?

Returns current wall-clock time.

What is Race Condition?

When multiple threads update shared data simultaneously and produce incorrect results.

What is Critical Section?

A section where only one thread executes at a time.

What is Reduction Clause?

Combines values safely from all threads.

Important CUDA Questions

What is a Kernel?

A function that runs on the GPU.

What Does `global` Mean?

The function runs on GPU and is called from CPU.

What is a Block?

Group of threads.

What is a Grid?

Group of blocks.

What is Host?

CPU memory.

What is Device?

GPU memory.

What Does cudaMalloc() Do?

Allocates memory on GPU.

What Does cudaMemcpy() Do?

Copies data between CPU and GPU.

What Does cudaFree() Do?

Frees GPU memory.

PART B: DEEP LEARNING

1. What is Artificial Intelligence (AI)?

Detailed Explanation

AI is the field of building systems that can perform tasks that normally require human intelligence.

Examples:

- Face recognition
- Voice assistants

- Recommendation systems

One-Line Answer

AI is the simulation of human intelligence by machines.

2. What is Machine Learning (ML)?

Detailed Explanation

ML is a subset of AI where systems learn patterns from data instead of being explicitly programmed.

Example

Spam detection learns from labeled emails.

One-Line Answer

Machine Learning enables systems to learn from data.

3. What is Deep Learning (DL)?

Detailed Explanation

Deep Learning is a subset of ML that uses neural networks with many hidden layers.

It automatically learns features from data.

Example

A deep learning model can learn to detect cats from images.

One-Line Answer

Deep Learning uses neural networks with multiple hidden layers.

Relationship Between AI, ML, and DL

AI $\supset$ ML $\supset$ DL

4. What is a Neural Network?

Detailed Explanation

A neural network is a collection of interconnected neurons arranged in layers:

- Input layer
- Hidden layers
- Output layer

Each neuron receives input, applies weights and bias, then produces output.

One-Line Answer

A neural network is a layered model that learns patterns from data.

5. What is a Neuron?

Detailed Explanation

A neuron performs:

1. Takes inputs.
2. Multiplies by weights.
3. Adds bias.
4. Applies activation function.
5. Produces output.

One-Line Answer

A neuron is the basic computational unit of a neural network.

6. What are Weights and Bias?

Weights

Determine the importance of each input.

Bias

Adds flexibility by shifting the output.

One-Line Answer

Weights control input importance, and bias shifts the output.

7. What is an Activation Function?

Detailed Explanation

It introduces non-linearity so the network can learn complex relationships.

Common Activation Functions

- ReLU
- Sigmoid
- Tanh
- Softmax

One-Line Answer

Activation functions help neural networks learn complex patterns.

8. ReLU

Formula: $\text{ReLU}(x) = \max(0, x)$

Advantages

- Simple
- Fast
- Reduces vanishing gradient

One-Line Answer

ReLU outputs 0 for negative values and the input itself for positive values.

9. Sigmoid

Outputs values between 0 and 1. Used for binary classification.

One-Line Answer

Sigmoid converts values into probabilities between 0 and 1.

10. Softmax

Converts outputs into probabilities that sum to 1. Used for multi-class classification.

One-Line Answer

Softmax converts outputs into class probabilities.

11. Forward Propagation

Computing outputs from inputs by passing data through the network.

One-Line Answer

Forward propagation calculates predictions from input data.

12. Loss Function

Measures prediction error.

Examples:

- Mean Squared Error
- Cross Entropy

One-Line Answer

Loss function measures how wrong the predictions are.

13. Backpropagation

Computes gradients and updates weights to reduce loss.

One-Line Answer

Backpropagation updates weights based on prediction error.

14. Gradient Descent

Optimization algorithm used to minimize loss.

One-Line Answer

Gradient descent updates weights to reduce error.

15. Learning Rate

Controls how much weights change.

- Too small → slow training
- Too large → unstable training

One-Line Answer

Learning rate controls the step size during weight updates.

16. Epoch, Batch Size, Iteration

Epoch

One complete pass through all training data.

Batch Size

Number of samples processed before updating weights.

Iteration

One weight update.

One-Line Answer

An epoch is one full pass through the dataset.

17. Overfitting

Model performs well on training data but poorly on new data.

Solutions

- Dropout
- More data
- Regularization
- Early stopping

One-Line Answer

Overfitting means memorizing training data instead of learning general patterns.

18. Underfitting

Model fails to learn the underlying pattern.

One-Line Answer

Underfitting means the model is too simple to learn the data.

19. Dropout

Randomly disables some neurons during training.

One-Line Answer

Dropout reduces overfitting by temporarily removing neurons.

20. CNN (Convolutional Neural Network)

Used for image processing.

Layers

- Convolution
- Activation
- Pooling
- Flatten
- Dense

Applications

- Face recognition
- Object detection
- Medical imaging

One-Line Answer

CNN is a neural network specialized for image data.

21. Convolution

Applying filters to detect features like edges and textures.

One-Line Answer

Convolution extracts features from images.

22. Pooling

Reduces feature map size.

Types

- Max Pooling
- Average Pooling

One-Line Answer

Pooling reduces dimensions while keeping important information.

23. Flatten

Converts 2D feature maps into a 1D vector.

One-Line Answer

Flatten converts feature maps into a vector.

24. Dense Layer

Fully connected layer where each neuron connects to all neurons in the previous layer.

One-Line Answer

Dense layer combines all extracted features for final prediction.

25. RNN

Used for sequential data like text and speech.

One-Line Answer

RNN processes data that has sequence information.

26. LSTM

A special type of RNN that remembers long-term dependencies.

One-Line Answer

LSTM is an improved RNN that can remember information for longer periods.

27. Transfer Learning

Using a pre-trained model and fine-tuning it for a new task.

One-Line Answer

Transfer learning reuses a previously trained model.

28. Data Augmentation

Creates new training samples by modifying existing data.

Examples:

- Rotation
- Flip
- Zoom

One-Line Answer

Data augmentation increases training data artificially.

29. Accuracy, Precision, Recall, F1 Score

Accuracy

Correct predictions / Total predictions

Precision

How many predicted positives are actually positive.

Recall

How many actual positives were correctly identified.

F1 Score

Balance between precision and recall.

30. TensorFlow, Keras, PyTorch

TensorFlow

Framework developed by Google.

Keras

High-level API for building neural networks.

PyTorch

Framework developed by Meta.